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Executive Summary

In this homework we took past PFD's for Refrigeration with Flash Tank Economizer and Two Stages of Compression and cyclohexane production that we simulated and change what is needed to observe the affects it has on cost and Production rate. We focus on seeing the affects increasing the stages of compression from two stages to three stages. Furthermore, we observe how a cooling tower is designed and how it can affect the calculation of the total operating cost, including the cost of raw materials, revenues from the product and credits for fuel. In the first problem, we take our previous PFD where we found the optimal interstage pressure for the cycle when we remove 3 MW of energy from a process gas stream in a chiller. Using this simulation, we add another stage of refrigeration and determine the optimal interstage pressures for the three-stage cycle. In the second problem, we take our previous PFD for cyclohexane production where we produced 100 KMTA, and increase it to 300 KMTA. Using this we add a cooling tower and calculate the total operating cost, including the cost of raw materials, revenues from the product and credits for fuel. We design an excel sheet that includes all the utilities used and the cost analysis for our entire process. This excel sheet included the shell and tube heat exchanger, compressors, pumps, and drums used.

Systems and Procedures

For our first problem we took our previous PFD for Refrigeration with Flash Tank Economizer and Two Stages of Compression. Using this PFD we first added another compressor in order to create a three stage refrigeration process. We created this process by having our first compressor go into mixer, the stream out entered the second compressor which then entered another mixer. The stream out entered our third compressor which was the inlet into the condenser. Now, that we had to add another mixer, we also had to add another separator. After specifying our new utilities added, we created a design specification to specify the chiller duty to be 3 MW or 3000 kJ/s. After getting our simulation to run and getting work at every one of our compressors, we did a sensitivity analysis for the optimization. In this analysis we varied our first and second compressors pressure, in order to find optimal interstage pressures for the three-stage cycle.

For our second problem, we look at our previous PFD for cyclohexane production where our target of production was 100 KMTA. Using the same design specification from the previous PFD, all we changed was the specification for our production target. Now, we had to use unit operations that correspond to actual equipment in the field. We first added a compressor which our top mixer outlet stream enters and the recycle gas stream exits which goes into the first heater. We then added a pump which our bottom mixer outlet stream enters and the liquid recycle stream exits which enters the first heater. Furthermore, we added a heater after our reactor which cools our temperature and then the stream out the heater enters a throttling valve. Our outlet stream from the bottom mixer enters a throttling valve which decreases our pressure and temperature. We made sure our reactor was at a discharge pressure of -10 atm, while all the heaters had a discharge pressure of -5 atm. Our compressor had an outlet pressure of 22.8 atm same as the inlet H₂ feed. Our pump operated at the same outlet pressure as our compressor. Once we got our simulation to

run, we connected a stream out our first heater into our second heater to further cool our effluent stream. Once this was added, we added another heater where cooling water was used to cool our stream going into the separator. Now that our process had all necessary utilities, equations were used for each type of equipment we used. Looking at our equations obtained from the Maxwell and Swift index used in appendix 1, you see the different types of equations used for purchase and installation cost for the shell and tube heat exchangers, pumps, compressors, and drums. Using the installation and purchase cost for every different type of equipment, we obtained the annualized cost for that specific equipment. Addition of every equipment's annualized cost, we see our total annualized cost for the entire cyclohexane production process of approximately \$6.2 million.

Results

1. Problem 4A – Refrigeration with Flash Tank Economizer and three Stages of

Compression

- i. With the given data, we were able to conduct a sensitivity analysis to find the optimal power required along with COP of the cycle and maximum thermodynamic COP of: **COP = 4.17 bar and Ideal COP = 4 bar, Compressor 1 = 579 kW and Compressor 2 = 237 kW, and Compressor 3 = 325 kW.**
- ii. We can conclude that the optimal pressure puts the COP close to the ideal COP.

2. Problem 4D – Cyclohexane Production

- i. With the given data, we were able to design a specification to meet the target ratio of hydrogen and liquid recycle to prepare a production target of 300 KMTA and our total annualized cost for each equipment. You can also see our total annualized cost for the entire process is about \$6.2 million
- ii. Additional Data:
 - i. PFD – See *Figure 2* in Appendix 2
 - ii. Material Stream Data – See *Table 2* in Appendix 3
 - iii. Heat Data – See *Table 3* in Appendix 3
 - iv. Capital Cost – See *Table 4* in Appendix 3

Conclusion

In conclusion we see how the number of stages of compression can affect the most optimum pressure and if we can obtain the necessary production rate. We compared out COP from the two-stage compression simulation versus the three stage compression COP. Furthermore, in our second problem we got to identify the actual equipment and utilities needed for the cyclohexane production to produce the necessary production target at high pressure while cooling our effluent stream. Once we had our entire PFD, we to help find the cost of all utilities in our process and calculate the total operating cost, including the cost of raw materials, revenues from the product and credits for fuel. Doing this by using the equations obtained from the Maxwell and Swift index, we understand how to build the best and most efficient process using the right utilities that won't drive you total annual cost up and instead you will save money and profit from the cyclohexane production.

Appendix 1

Equations and Formulas

Shell and Tube Heat Exchangers:

$$\text{Purchase Cost(2010)} = 28,000 + 54 * A^{1.2}$$

$$\text{Purchase Cost(2020)} = \text{Purchase Cost(2010)} + \frac{\text{index(2020)}}{\text{index(2010)}}$$

$$\text{Where: } \text{index(2010)} = 1500$$

$$\text{To obtain 'A': } Q = F_t * A * U * \Delta T_{ln}$$

$$\text{Where: } Q = \text{Obtained from Aspen Simulation}$$

$$F_t = 0.8$$

$$U = 50 \frac{\text{Btu}}{\text{hr} * \text{ft}^2 * ^\circ\text{F}}$$

$$\Delta T_{ln} = \frac{\Delta T_1 - \Delta T_2}{\ln (\Delta T_1 / \Delta T_2)}$$

$$\text{Installation Factor} = 3.5$$

$$\text{Installation Cost} = 3.5 * \text{Purchase Cost(2020)}$$

$$\text{Annualized Cost} = \frac{4}{3} * \text{Installation Cost}$$

Pumps:

$$\text{Purchase Cost(2010)} = 8,000 + 240 * Q^{0.9}$$

$$\text{Purchase Cost(2020)} = \text{Purchase Cost(2010)} + \frac{\text{index(2020)}}{\text{index(2010)}}$$

$$\text{Where: } \text{index(2010)} = 1500$$

$$Q = \text{Obtained from Aspen Simulation in } \frac{L}{s}$$

$$\text{Installation Factor} = 4$$

$$\text{Installation Cost} = 4 * \text{Purchase Cost(2020)}$$

$$\text{Annualized Cost} = \frac{4}{3} * \text{Installation Cost}$$

Compressors:

$$\text{Purchase Cost(2010)} = 500,000 + 20,000 * W^{0.6}$$

$$\text{Purchase Cost(2020)} = \text{Purchase Cost(2010)} + \frac{\text{index(2020)}}{\text{index(2010)}}$$

$$\text{Where: index(2010)} = 1500$$

W= Obtained from Aspen Simulation in Kw.

$$\text{Installation Factor} = 2.5$$

$$\text{Installation Cost} = 2.5 * \text{Purchase Cost(2020)}$$

$$\text{Annualized Cost} = \frac{4}{3} * \text{Installation Cost}$$

Drums:

$$\text{Assumptions: } \frac{L}{D} = 2, \text{ Residence Time} = 5 \text{ minutes}$$

$$\text{Purchase Cost(2010)} = 11,600 + 34 * M^{0.85}$$

$$\text{Purchase Cost(2020)} = \text{Purchase Cost(2010)} + \frac{\text{index(2020)}}{\text{index(2010)}}$$

$$\text{Where: index(2010)} = 1500$$

$$\text{Where: } M = (\pi * D * L) * \text{Thickness} * \rho_{steel}$$

$$\rho_{steel} = 8000 \frac{kg}{m^3}$$

$$\text{Thickness} = \frac{P_i D_i}{2 * S * E - 1.2 * P_i}$$

$$\text{Where: } E = \text{Welded Efficiency} = 1$$

$$S_{steel} = \text{Stress} = 12.9 \text{ Ksi}$$

$$P_i = \text{Internal Operating Pressure From Aspen Simulation plus 10\% safety factor}$$

$$D_i = \text{Internal Diameter}$$

$$\text{Where: } D_i = \frac{2 * (V_{vessel})^{1/3}}{\pi}$$

Installation Factor= 4

Installation Cost= 4* Purchase Cost(2020)

Annualized Cost= $\frac{4}{3}$ * Installation Cost

Appendix 2

PFD

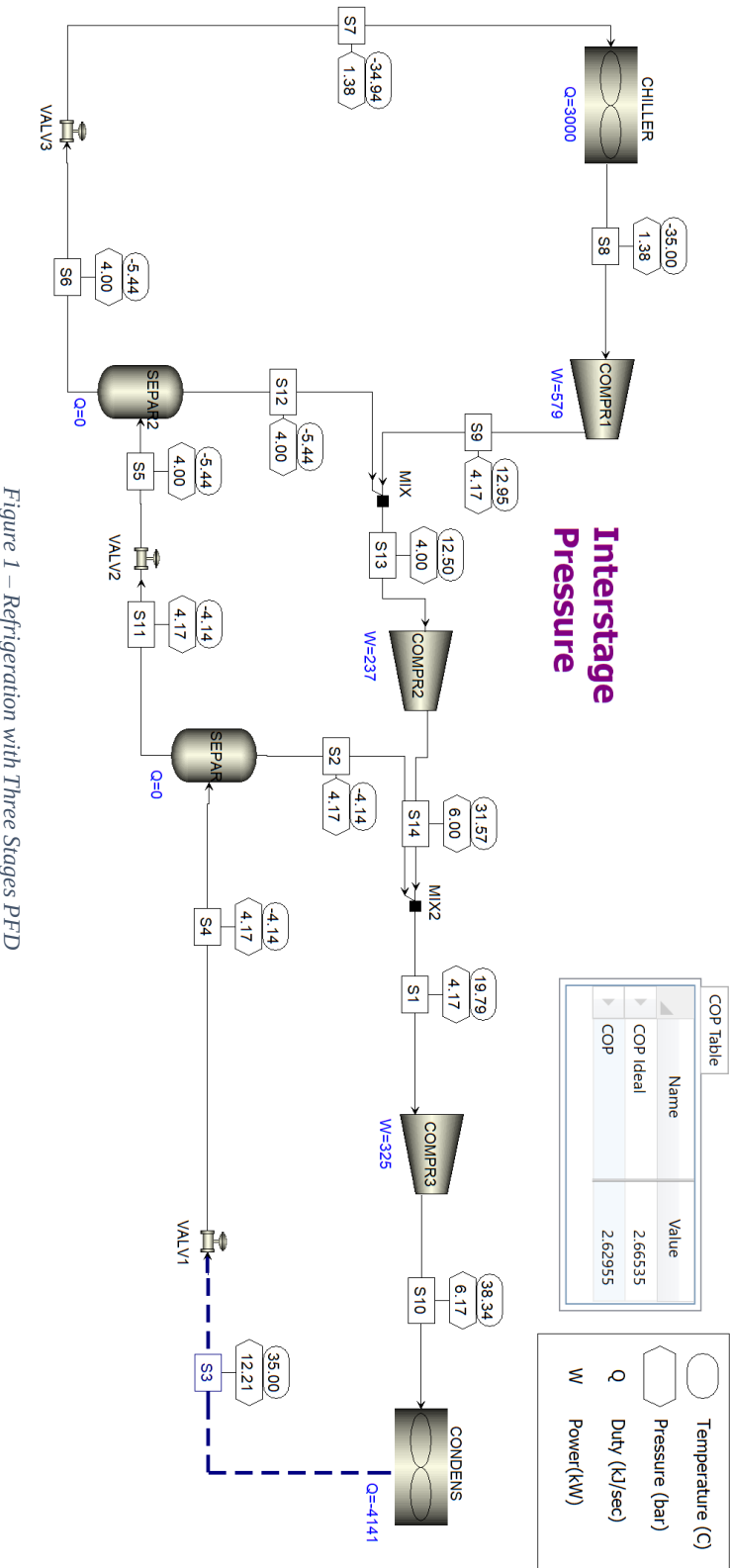


Figure 1 – Refrigeration with Three Stages PFD

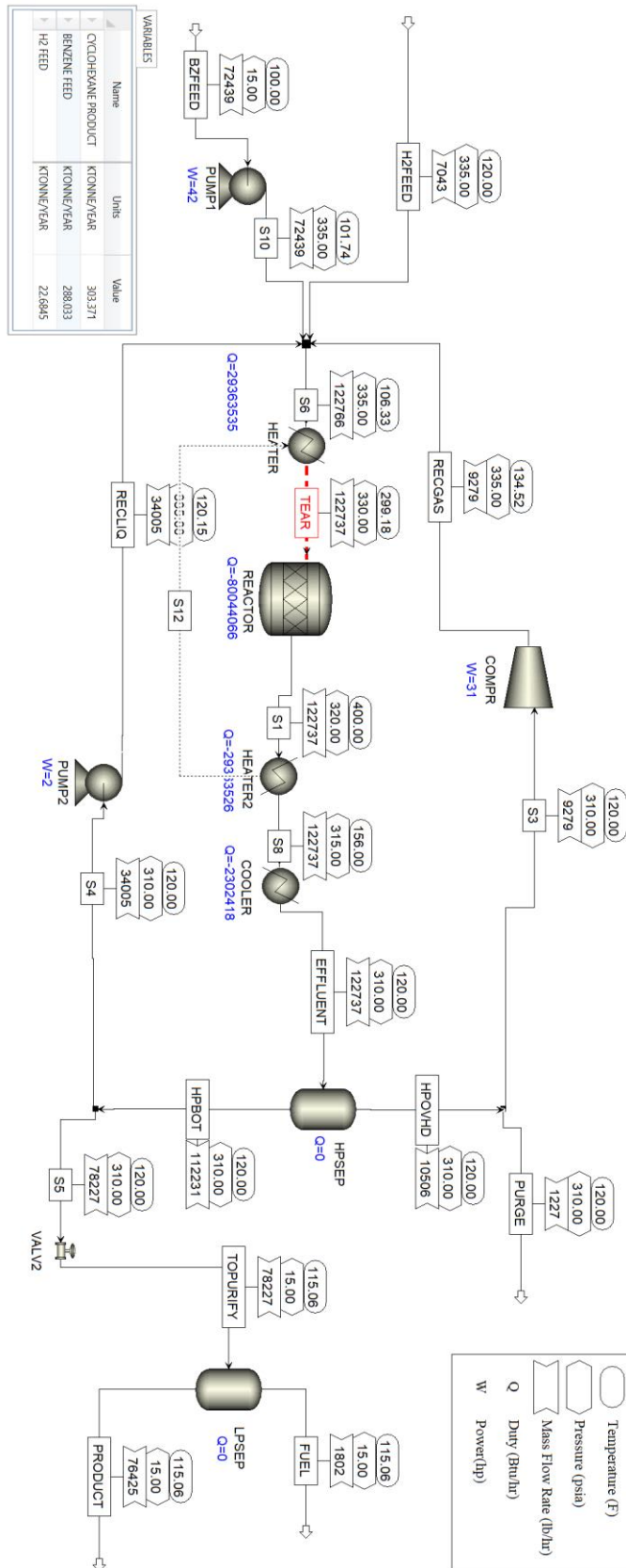


Figure 2 – Cyclohexane Production PFD

Appendix 3

Energy/Material Balance

Stream Name Description	Units	Material															
		S1	S10	S11	S12	S13	S14	S2	S3	S4	S5	S6	S7	S8	S9		
From		MIX2	COMPR3	SEPAR	SEPAR2	MIX	COMPR2	SEPAR	CONDENS	VALV1	VALV2	SEPAR2	VALV3	CHILLER	COMPR1		
To		COMPR3	CONDENS	VALV2	MIX	COMPR2	MIX2	MIX2	VALV1	SEPAR	SEPAR2	VALV3	CHILLER	COMPR1	MIX		
MIXED Substream																	
Phase		Vapor	Vapor	Liquid	Vapor	Vapor	Vapor	Vapor	Liquid	Mixed	Mixed	Liquid	Mixed	Vapor	Vapor		
Temperature	C	19.7913	38.3445	-4.14061	-5.44167	12.4981	31.573	-4.14061	35	-4.14061	-5.44167	-5.44167	-34.9391	-35	12.9522		
Pressure	bar	4.16842	6.16842	4.16842	4	4	6	4.16842	12.2062	4.16842	4	4	1.38	1.37656	4.16842		
Molar Vapor Fraction		1	1	0	1	1	1	1	0	0.275955	0.00840397	0	0.167311	1	1		
Molar Liquid Fraction		0	0	1	0	0	0	0	1	0.724045	0.991596	1	0.832689	0	0		
Mass Entropy	kJ/kg-K	-6.42791	-6.4061	-7.99219	-6.56957	-6.46186	-6.4394	-6.57083	-7.63197	-7.59996	-7.99205	-8.00411	-7.98537	-6.52768	-6.46813		
Average MW		44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965		
Mole Flows	kmol/hr	982.718	982.718	711.531	5.97969	711.531	711.531	271.186	982.718	982.718	711.531	705.552	705.552	705.552	705.552		
Mass Flows	kg/hr	43334.4	43334.4	31376.1	263.684	31376.1	31376.1	11958.4	43334.4	43334.4	31376.1	31112.4	31112.4	31112.4	31112.4		
Volume Flow	cum/hr	5307.95	3731.41	58.6945	30.1445	3895.68	2706.34	1313.98	91.0473	1372.67	88.1566	58.0121	1667.71	9720.26	3700.19		
Molar Enthalpy	kJ/kmol	-105564	-104373	-124195	-107403	-106089	-104889	-107334	-119542	-119542	-124195	-124337	-124337	-109030	-106078		
Mass Enthalpy	kJ/kg	-2393.93	-2366.92	-2816.44	-2435.63	-2405.83	-2378.63	-2434.07	-2710.92	-2710.92	-2816.44	-2819.66	-2819.66	-2472.53	-2405.58		
Enthalpy Flow	kJ/sec	-28816.6	-28491.4	-24546.8	-178.398	-20968.2	-20731.1	-8085.43	-32632.3	-32632.3	-24546.8	-24368.4	-24368.4	-21368.4	-20789.8		
Molar Entropy	kJ/kmol-K	-283.448	-282.487	-352.428	-289.695	-284.945	-283.955	-289.751	-336.543	-335.132	-352.422	-352.953	-352.127	-287.848	-285.222		
Mass Entropy	kJ/kg-K	-6.42791	-6.4061		-6.56957	-6.46186	-6.4394	-6.57083	-6.57083	-6.57083	-6.56957		-6.5278	-6.52768	-6.46813		

Table 1 – Refrigeration with Three Stages of Compression Material Balance

Material													
Stream Name	Units	BZFEED	EFFLUENT	FUEL	H2FEED	HPBOT	HPOVHD	PRODUCT	PURGE	REGAS	RELIQ	TEAR	TOPURIFY
From			COOLER	LPSEP		HPSEP	HPSEP			COMPR		HEATER	
To		PUMP1	HPSEP		MIX1	MIX3				MIX1	MIX1	REACTOR	LPSEP
Stream Class		CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN
Temperature	F	100	120	115.061	120	120	120	115.061	120	134.521	120.152	299.176	115.061
Pressure	psia	15	310	15	335	310	310	15	310	335	335	330	15
Molar Density	lbmol/cuft	0.686671	0.122001	0.00244952	0.0531551	0.566225	0.0499393	0.558444	0.0499393	0.0525874	0.566164	0.041104	0.0437681
Average MW		78.1136	57.7678	36.2546	2.42641	81.5425	14.0396	84.0169	14.0396	81.5425	81.5425	25.0179	81.5425
Mole Flows	lbmol/hr	927.36	2124.67	49.7002	2902.64	1376.35	748.313	909.636	87.3722	660.941	417.017	4905.98	959.337
H2	lbmol/hr	0	266.194	2.8964	2830.07	4.18636	262.008	0.0215424	30.5917	231.416	1.26841	3061.43	2.91795
N2	lbmol/hr	0	101.888	2.96944	14.5132	4.35305	97.5346	0.0646915	11.388	86.1466	1.31892	101.888	3.03414
Cl	lbmol/hr	0	418.076	29.011	58.0527	44.1142	373.962	1.73719	43.6634	330.299	13.366	401.372	30.7482
BZ	lbmol/hr	927.36	1.85607	0.0219139	0	1.83445	0.0216134	1.25672	0.00252356	0.0190899	0.555815	928.034	1.27864
CH	lbmol/hr	0	1336.65	14.8014	0	1321.87	14.7874	906.556	1.72655	13.0608	400.508	413.259	921.358
Mass Flows	lb/hr	72439.4	122737	1801.86	7042.97	112231	10506.1	76424.8	1226.67	9279.38	34004.6	122737	78226.7
H2	lb/hr	0	536.615	5.8388	5705.08	8.43919	528.176	0.0434269	61.6692	466.507	2.55696	6171.48	5.88223
N2	lb/hr	0	2854.23	83.1845	406.565	121.944	2732.28	1.81223	319.018	2413.27	36.9475	2854.23	84.9967
Cl	lb/hr	0	6707.1	465.416	931.326	707.713	5999.38	27.8693	700.481	5298.9	214.428	6439.11	493.286
BZ	lb/hr	72439.4	144.984	1.71177	0	143.296	1.6883	98.1673	0.197124	1.49118	43.4168	72492.1	99.8791
CH	lb/hr	0	112494	1245.71	0	111250	1244.52	76296.9	145.309	1099.21	33707.2	34780.4	77542.7
Mass Fractions													
H2		0	0.00437206	0.00324043	0.810039	7.52E-05	0.0502735	5.68E-07	0.0502735	0.0502735	7.52E-05	0.050282	7.52E-05
N2		0	0.0232548	0.0461659	0.0577263	0.00108654	0.260068	2.37E-05	0.260068	0.260068	0.00108654	0.0232548	0.00108654
Cl		0	0.0546459	0.258298	0.132235	0.00630585	0.57104	0.000364663	0.57104	0.57104	0.00630585	0.0524625	0.00630585
BZ		1	0.00118126	0.000950003	0	0.00127679	0.000160698	0.0012845	0.000160698	0.000160698	0.00127679	0.590628	0.00127679
CH		0	0.916546	0.691346	0	0.991256	0.118458	0.998327	0.118458	0.118458	0.991256	0.283373	0.991256
Volume Flow	cuft/hr	1350.51	17415.2	20289.7	54606.9	2430.75	14984.5	1628.88	1749.57	12568.4	736.566	119355	21918.6
Molar Enthalpy	Btu/lbmol	21902	-47349	-34060.9	-334.286	-63937.2	-16838.8	-65569.6	-16638.8	-16718.9	-63926.3	2264.36	-63937.2
Mass Enthalpy	Btu/lb	280.386	-819.644	-939.493	-137.77	-784.097	-1199.38	-780.433	-1199.38	-1190.83	-783.963	90.5097	-784.097

Table 2 – Cyclohexane Production Material Balance

Heat	
Stream Name	S12
QCALC Btu/hr	29363535
TBEGIN F	400
TEND F	156

Table 3 – Cyclohexane Production Heat

Cyclohexane Production Cost Estimate						
EQUIPMENT LIST: CAPITAL COSTS						
	Project:	Assignment #4	Chemical Engineers:	Khalil Shehaded	Esteban Vazquez-Elizondo	
	Location:	Chemical Engineering Department	Date:	10/22/2019		
Equipment Class	ASPEN BlockID	Service Description	Tag	Material Cost	Installed Cost	Annual Cost, \$/yr
Distillation						
Drums						
Vertical Tank	HPSEP	High Pressure Separator Drum	V-1	\$340,443.00	\$1,361,772.00	\$1,815,696.00
Vertical Tank	LPSEP	Low Pressure Separator Drum	V-2	\$60,763.70	\$243,054.80	\$324,073.07
Heat Exchangers						
S&T Exchanger	HEATER	Feed-Recycle Exchanger	E-1	\$310,735.00	\$1,087,572.50	\$1,450,096.67
S&T Exchanger	HEATER2	Effluent Cooler	E-2	\$39,957.50	\$139,851.25	\$186,468.33
S&T Exchanger	COOLER	Effluent Cooler 2	E-3	\$42,816.50	\$149,857.75	\$199,810.33
Air Coolers						
Pumps						
Pump	PUMP1	Benzene Pump	P-1	\$8,157.32	\$32,629.28	\$43,505.71
Pump	PUMP2	Recycle Pump	P-2	\$13,324.80	\$53,299.20	\$71,065.60
Turbines						
Compressors						
Compressor	COMPR	Recycle Gas Compressor	C-1	\$632,475.00	\$1,581,187.50	\$2,108,250.00
Reactors						

Table 4 – Cyclohexane Production Cost Estimate

References

Marshall and Swift Cost Index, M&S